

Target-conditioned refinement of future-blind longitudinal glioma segmentation-change maps across independent cohorts

Supplementary Information

[AUTHOR_LIST_REQUIRED]

V1 was frozen before revision. Table S1 records its files. Development P0 was fold-specific out-of-fold inference, whereas the independent internal and RHUH P0 maps were equal-weight ensembles of five frozen checkpoints. Table S2 provides the exact clinical-metadata linkage and missingness without imputation.

Fixed, Naive, EIA and PCC definitions follow LOCKED_COMPARATOR_METHOD_DEFINITIONS.md. EIA and target-volume top-k analyses require the target and are retrospective oracle-style controls. No secondary pairwise inferential tests were introduced.

Development mechanism summaries (Table S5), imperfect-guidance analyses (Table S6) and no-smoothing robustness (Table S7) use the frozen development authorities. Development top-k outcomes are target-volume-matched oracle-assisted localization metrics, not Dice@0.5.

Internal and RHUH secondary metrics are reported in Tables S3 and S4. Oracle-assisted controls appear separately in Tables S8 and S9. Table S10 accounts for all scientific denominators and failures. Spatial analyses remain exploratory and are not used for biological or causal claims.

Table S1. Authority and frozen V1 provenance. Path, size and SHA-256 for every frozen V1 package file.

Table S2. Extended locked-cohort clinical metadata. Exact patient-ID linkage to official MU-Glioma-Post and RHUH clinical metadata; missing and invalid intervals are retained without imputation.

Table S3. Internal secondary metrics. Descriptive patient-level summaries and locked bootstrap intervals; no new P values.

Table S4. RHUH secondary metrics. Descriptive patient-level summaries and locked bootstrap intervals; no new P values.

Table S5. Development mechanism and method summaries. Development-only results; target-volume-matched metrics are oracle-assisted.

Table S6. Imperfect-guidance robustness. Frozen perturbation summaries.

Table S7. No-smoothing robustness. Frozen condition-by-method summaries.

Table S8. Internal oracle-assisted controls. Target-volume-matched top-k summaries.

Table S9. RHUH oracle-assisted controls. Target-volume-matched top-k summaries.

Table S10. Failure accounting. Fixed denominators and scientific failures.

Table S1. Authority and frozen V1 provenance

Part 1 of 1

relative path	size bytes	sha256	frozen
01_PCC_SCIENTIFIC_REPORTS_SUBMISSION_REA DY.docx	1104288	39348ae448c5b1e1793c8f9a525547996fb6fe9bf2f0381 1696637c7e5577c7b	true
02_PCC_SCIENTIFIC_REPORTS_SUBMISSION_REA DY.pdf	1263868	127996ce1e2fcce21e77fafc676ba32cc0c0709b5e1547f dce1b0661be0acc2d	true
03_PCC_SCIENTIFIC_REPORTS_SUPPLEMENTARY _INFORMATION.docx	48864	e6195a594ff0b8ca019df7b58e83c3e7c045645b1b4d01 4d67e5e4a07a86902a	true
04_PCC_SCIENTIFIC_REPORTS_SUPPLEMENTARY _INFORMATION.pdf	31541	1586c8b7e8f9c8bcf0a41e41438503c719a3f88ab24fb0 3b1b3c517aafa4b7a5	true
05_PCC_SCIENTIFIC_REPORTS_COVER_LETTER.d ocx	38072	8a4f0e0361d1635a97b8b7ca29d0fbee3661fe81790367 531f3b521f2a1924ad	true
AUDITS/CLAIM_2024_COMPLIANCE.csv	5546	c319a4f67b7e993ea7eb019012717574252f47bfe40514 359eeaf4060d484461	true
AUDITS/CLAIM_2024_COMPLIANCE_REPORT.md	612	e8333e4c6d1c525b07d491802f9503fcb04dd15458411 c2ddf4946943bf83fc8	true
AUDITS/CLAIM_TO_REFERENCE_MATRIX.csv	633	f1eeee344fc4f4b8d1340ba130f64551737aa4ac673b25 779235ccc71a35dee0	true
AUDITS/FINAL_REVIEWER_ISSUE_MATRIX.csv	533	6c76b5c07e92de929c152f30b72fe34d4dda75f6cde9dd cc4a628cb1dafede01	true
AUDITS/MANUSCRIPT_CITATION_CLAIM_AUDIT.csv	964	012663d355aaee38ed7211286c457a2df8562bd58e75a 27260164e43e1e1a0e4	true
AUDITS/MANUSCRIPT_NUMERIC_CLAIM_AUDIT.csv	18583	d7ffc72c95012ffcfa1fd9fadbccb41cbb962ef0fbeb7e07a 68495a5ed5d29e	true
AUDITS/MANUSCRIPT_RELEASE_TEST_REPORT.m d	185	a4a59cd0c3022e8b4d14356455fe48c5802f65518842f2 1afdcfbf0afa11d0ec	true
AUDITS/MANUSCRIPT_RELEASE_TEST_RESULTS.c sv	2700	87a82aa6bb72bba99b8009b242864e98fb61fa3344e8b 4815fce1b5669ad0dba	true
AUDITS/PCC_MANUSCRIPT_AUTHORITATIVE_EVID ENCE_REGISTRY.csv	4123	ef5ad0c136623724b56ccd027e55432bb2bb8b2ac97d2 bd39a3f3ae0392891dc	true
AUDITS/PCC_MANUSCRIPT_AUTHORITY_REPORT. md	1248	34629156e34d357fac6db1a2bb4c9542488a04b4cdf5fb 9764ab8dc3e6ab8919	true
AUDITS/PCC_SUPERSEDED_AND_HISTORICAL_RE SULTS_REGISTRY.csv	1011	5f862bd8d80af2f5d9693e63555f8481a9e3bae39f04735 66722dd006e7c4624	true
AUDITS/PROBAST_AI_INTERNAL_RISK_AUDIT.md	983	c56133eb8a7901f36edc955e4753d8983fa6300d26174 ec2046d6d8fdc165505	true
AUDITS/REFERENCE_MASTER_LEDGER.csv	4409	2fb634b189a153da70a1875516b2a0d4ba6d0e361096b c3743029ad960389fab	true
AUDITS/REFERENCE_VERIFICATION_REPORT.md	445	d996b9f96f80c9261f69347e5dffa47c3472cbac678d9ea 48405f1a09e580e2d	true

relative path	size bytes	sha256	frozen
AUDITS/SCIENTIFIC_REPORTS_COMPLIANCE_MATRIX.csv	1097	a36fe1a57a39272c89aa7b97f5b3c4cee7cbb9cb35abaa cbcec31b9f20bbd956	true
AUDITS/SCIENTIFIC_REPORTS_CURRENT_REQUIREMENTS.md	2244	1e21ce96e16f5c4ff6113146d74e2f09f218f6277fb4ed35 3a1ff3b5ac2398f0	true
AUDITS/TRIPOD_AI_APPLICABILITY_AUDIT.md	610	2b0883498b8716b4b7916ffc8c13aadd8b15b66ca482 38675e05e668741bd58	true
AUDITS/TRIPOD_AI_COMPLIANCE_IF_APPLICABLE.csv	2443	5e58ee484b63064af6c8510a76c0b0d389fe77ffd607a8f b2dc60dc1bc0a7026	true
AVAILABILITY/CODE_AVAILABILITY_DRAFT.md	379	55cd37e8f6ff93351d054727c8239a37260a8bb8abd61d 1510104a618339af0c	true
AVAILABILITY/CODE_RELEASE_PLAN.md	521	c310ef82be6ae62a4f77a8a50ee9a4d306ff9b935170d5f 7c9bf6c74d2c7d8bb	true
AVAILABILITY/DATA_AVAILABILITY_DRAFT.md	631	91f49e06f9c68f342edf927233500f34954827d20997efb 4764e5906958e82ed	true
ETHICS/AI_ASSISTANCE_DISCLOSURE_DRAFT.md	559	d56522b9e169dbf411b4e04a50ddd27b3529d431bac71 2823ca72e8cefade689	true
ETHICS/ETHICS_MANUSCRIPT_TEXT.md	610	2cf52d739253950cfa632ce3ffeb8d6b0b68aa27496250f 6f545b75013337f38	true
ETHICS/ETHICS_SOURCE_AUDIT.md	1206	175bbb1f99d965da22cc17876a2c5d2b9915063d43dd9 52b587bd0dd4afb1af	true
FIGURES/Figure_1_study_design.png	142843	81fe3e73be228315042e0fdb712ff0075d70fa393d341d 21bd6e3202176b4b7	true
FIGURES/Figure_1_study_design.svg	2240	92ab74b87a9c2b18e2691c15014a360109c64be266c2 30b5d3cbfdd074c6a163	true
FIGURES/Figure_2_internal_confirmatory.png	558333	f237b42795b9c885be68f6a44fb75b38b8a7bb27267a8e 4069744fd4d640d373	true
FIGURES/Figure_2_internal_confirmatory.svg	39130	b9afc5933088d4517ff87c46add985cfb2b66fc6649482c b023fb8dd9c8bfaad	true
FIGURES/Figure_3_full_pcc_trajectories.png	103613	e9746fe784c7917d5c4835dacb74f46395d72359dfbca8 80238e0fb5d5e63e84	true
FIGURES/Figure_3_full_pcc_trajectories.svg	3464	618d886aa618d98755f83ffc5230f9ab981928a3087a6c 44fae66659db29f9be	true
FIGURES/Figure_4_rhuh_external.png	239303	9579943e325acb1fa7d645dd25694f1f353639a1f94d27 001593a350a4544057	true
FIGURES/Figure_4_rhuh_external.svg	14615	5885835e790082ebedab16fac60c62badf4b34faa893ad c2d892d351e3c29450	true
FIGURES/Figure_5_development_controls.png	91386	a09ea6bc5fb301190ba11bc233444506c62e141cab357 9779f35b219066787be	true
FIGURES/Figure_5_development_controls.svg	1898	b77e06eb6c665da98d7a1ef39fac49d600e6bd397c44e 44dc523ffc47623ec4d	true
FINAL_MANUSCRIPT_RELEASE_REPORT.md	792	07e4cda320b2f2fcdff45b3a10f024f3165410646b0b148 44f5fe3ea7df78766	true

relative path	size bytes	sha256	frozen
FINAL_SUBMISSION_READINESS_ASSESSMENT.md	1028	9046d0b9818e84de04a6e646108252128585f9db1c562c9166c5fc2515419a74	true
MANUSCRIPT_CHANGELOG.md	478	6200197ab73fc093c1a60111e4433c8697df9707cc8014a119b2b544d3254e2d	true
PACKAGE_FILE_MANIFEST.csv	6353	86eea5d88dbb1f8ec0f82d571a3b91756884eac20f6955bf81c26c37afb4242d	true
PACKAGE_VALIDATION_REPORT.md	396	bdeddc632a520c870ae077d84dcf7f126909ed59ebe92b72aee1bdbeea5b0000	true
SUBMISSION_ACTIONS_REQUIRED.md	1196	02828720045080078fbcfda33904b0b758b03ff401476725c206484a15d81332	true
TABLES/Supplementary_Table_S2_internal_secondary.csv	12571	f0432bc72cb628f67785fda83701b6fcc5f7889f255009fc8e7b5e4b79da63fa	true
TABLES/Supplementary_Table_S3_rhuh_secondary.csv	13854	0fc8f49d6697bcf35656165d1ff470ba002aa27da5ed80c510ca167ec0215ac3	true
TABLES/Supplementary_Table_S4_no_smoothing_robustness.csv	15680	f931db7717b2a4f76a8268286c79e6bae866add81c03e2c4fa9d4437c7d47d53	true
TABLES/Supplementary_Table_S7a_internal_oracle.csv	2111	29685faf2d3109c1c96ae74fd5af9c4d4b3d403715adcb78b358f3ea425ab6fa	true
TABLES/Supplementary_Table_S7b_rhuh_oracle.csv	3464	d6999dfcdaacf30745ae870cf4709166b9595db36e3293b0f0d5aa74525d20ab	true
TABLES/Table_1_cohort_design.csv	243	7106f1f9dd1bf4527d9a6f181ac99f4bdf99393acabe216bb20b3033c7b1d8b1	true
TABLES/Table_2_confirmatory_statistics.csv	892	686cc82552a94f6596f8e6a3c5c49aae7964be7f5e81bf3167e9a7609aa13c3c	true

Table S2. Extended locked-cohort clinical metadata

Part 1 of 2

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Development	PatientID_0003	PatientID_0003_T1_to_T2_t1c	57	Female	GBM	4	56
Development	PatientID_0005	PatientID_0005_T3_to_T4_t1c	49	Male	GBM	4	43
Development	PatientID_0006	PatientID_0006_T2_to_T4_t1c	60	Male	GBM	4	42
Development	PatientID_0007	PatientID_0007_T2_to_T3_t1c	79	Male	GBM	4	
Development	PatientID_0008	PatientID_0008_T4_to_T6_t1c	50	Male	GBM	4	113
Development	PatientID_0010	PatientID_0010_T1_to_T4_t1c	61	Male	GBM	4	133
Development	PatientID_0011	PatientID_0011_T1_to_T2_t1c	67	Male	GBM	4	70
Development	PatientID_0012	PatientID_0012_T2_to_T3_t1c	53	Male	GBM	4	77
Development	PatientID_0013	PatientID_0013_T1_to_T2_t1c	46	Female	GBM	4	98
Development	PatientID_0014	PatientID_0014_T1_to_T2_t1c	53	Female	GBM	4	42
Development	PatientID_0018	PatientID_0018_T1_to_T2_t1c	53	Male	GBM	4	92
Development	PatientID_0019	PatientID_0019_T4_to_T5_t1c	54	Female	GBM	4	49
Development	PatientID_0020	PatientID_0020_T1_to_T2_t1c	56	Male	GBM	4	76
Development	PatientID_0021	PatientID_0021_T2_to_T3_t1c	58	Male	GBM	4	92
Development	PatientID_0022	PatientID_0022_T1_to_T2_t1c	53	Male	GBM	4	52
Development	PatientID_0024	PatientID_0024_T2_to_T3_t1c	78	Female	GBM	4	123
Development	PatientID_0025	PatientID_0025_T1_to_T2_t1c	58	Female	GBM	4	114
Development	PatientID_0026	PatientID_0026_T1_to_T2_t1c	57	Male	GBM	4	111
Development	PatientID_0029	PatientID_0029_T1_to_T3_t1c	69	Male	GBM	4	183

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Development	PatientID_0030	PatientID_0030_T1_to_T3_t1c	69	Female	GBM	4	318
Development	PatientID_0031	PatientID_0031_T2_to_T3_t1c	53	Male	GBM	4	78
Development	PatientID_0032	PatientID_0032_T1_to_T2_t1c	60	Female	GBM	4	63
Development	PatientID_0033	PatientID_0033_T1_to_T2_t1c	53	Male	GBM	4	71
Development	PatientID_0034	PatientID_0034_T1_to_T2_t1c	53	Male	GBM	4	62
Development	PatientID_0035	PatientID_0035_T1_to_T2_t1c	54	Male	GBM	4	91
Development	PatientID_0036	PatientID_0036_T1_to_T2_t1c	55	Female	GBM	4	78
Development	PatientID_0037	PatientID_0037_T1_to_T2_t1c	87	Male	GBM	4	105
Development	PatientID_0038	PatientID_0038_T1_to_T2_t1c	53	Male	GBM	4	92
Development	PatientID_0039	PatientID_0039_T1_to_T2_t1c	66	Female	GBM	4	98
Development	PatientID_0041	PatientID_0041_T1_to_T2_t1c	74	Female	GBM	4	91
Development	PatientID_0044	PatientID_0044_T1_to_T2_t1c	50	Female	GBM	4	65
Development	PatientID_0045	PatientID_0045_T1_to_T2_t1c	64	Male	GBM	4	70
Development	PatientID_0051	PatientID_0051_T1_to_T3_t1c	58	Female	GBM	4	90
Development	PatientID_0052	PatientID_0052_T1_to_T2_t1c	58	Female	GBM	4	70
Development	PatientID_0053	PatientID_0053_T1_to_T3_t1c	37	Female	GBM	4	151
Development	PatientID_0054	PatientID_0054_T1_to_T2_t1c	65	Female	GBM	4	77
Development	PatientID_0055	PatientID_0055_T1_to_T4_t1c	63	Male	GBM	4	182
Development	PatientID_0059	PatientID_0059_T1_to_T2_t1c	52	Female	GBM	4	1109
Development	PatientID_0060	PatientID_0060_T1_to_T2_t1c	65	Male	GBM	4	3
Development	PatientID_0062	PatientID_0062_T1_to_T2_t1c	65	Male	GBM	4	3

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Independent internal	PatientID_0064	PatientID_0064_T1_to_T2_t1c	64	Male	GBM	4	84
Independent internal	PatientID_0065	PatientID_0065_T1_to_T2_t1c	63	Female	GBM	4	14
Independent internal	PatientID_0066	PatientID_0066_T1_to_T2_t1c	53	Female	GBM	4	63
Independent internal	PatientID_0067	PatientID_0067_T1_to_T2_t1c	42	Male	GBM	4	103
Independent internal	PatientID_0068	PatientID_0068_T1_to_T2_t1c	78	Female	GBM	4	77
Independent internal	PatientID_0069	PatientID_0069_T2_to_T3_t1c	72	Male	GBM	4	51
Independent internal	PatientID_0070	PatientID_0070_T1_to_T2_t1c	40	Male	GBM	4	3
Independent internal	PatientID_0072	PatientID_0072_T1_to_T2_t1c	54	Male	GBM	4	103
Independent internal	PatientID_0073	PatientID_0073_T1_to_T2_t1c	63	Male	GBM	4	43
Independent internal	PatientID_0074	PatientID_0074_T1_to_T2_t1c	44	Male	GBM	4	81
Independent internal	PatientID_0076	PatientID_0076_T1_to_T2_t1c	73	Male	GBM	4	41
Independent internal	PatientID_0078	PatientID_0078_T1_to_T2_t1c	65	Female	GBM	4	103
Independent internal	PatientID_0079	PatientID_0079_T1_to_T2_t1c	69	Male	GBM	4	25
Independent internal	PatientID_0080	PatientID_0080_T1_to_T2_t1c	78	Male	GBM	4	31
Independent internal	PatientID_0081	PatientID_0081_T1_to_T2_t1c	50	Male	GBM	4	
Independent internal	PatientID_0082	PatientID_0082_T1_to_T2_t1c	57	Male	GBM	4	128
Independent internal	PatientID_0083	PatientID_0083_T1_to_T2_t1c	71	Male	GBM	4	39
Independent internal	PatientID_0084	PatientID_0084_T1_to_T2_t1c	69	Male	GBM	4	101
Independent internal	PatientID_0085	PatientID_0085_T1_to_T2_t1c	37	Male	GBM	4	73
Independent internal	PatientID_0087	PatientID_0087_T1_to_T2_t1c	61	Female	GBM	4	129
Independent internal	PatientID_0088	PatientID_0088_T1_to_T2_t1c	78	Female	GBM	4	11

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Independent internal	PatientID_0089	PatientID_0089_T1_to_T2_t1c	48	Male	GBM	4	89
Independent internal	PatientID_0090	PatientID_0090_T1_to_T2_t1c	69	Male	GBM	4	91
Independent internal	PatientID_0091	PatientID_0091_T1_to_T2_t1c	76	Male	GBM	4	33
Independent internal	PatientID_0092	PatientID_0092_T1_to_T2_t1c	54	Male	GBM	4	35
Independent internal	PatientID_0093	PatientID_0093_T1_to_T2_t1c	61	Female	GBM	4	63
Independent internal	PatientID_0094	PatientID_0094_T1_to_T2_t1c	52	Female	GBM	4	91
Independent internal	PatientID_0095	PatientID_0095_T1_to_T2_t1c	59	Male	GBM	4	80
Independent internal	PatientID_0097	PatientID_0097_T1_to_T2_t1c	67	Male	GBM	4	69
Independent internal	PatientID_0099	PatientID_0099_T1_to_T2_t1c	63	Male	GBM	4	49
Independent internal	PatientID_0103	PatientID_0103_T1_to_T2_t1c	46	Female	GBM	4	68
Independent internal	PatientID_0105	PatientID_0105_T1_to_T2_t1c	63	Male	GBM	4	91
Independent internal	PatientID_0109	PatientID_0109_T1_to_T2_t1c	59	Female	GBM	4	31
Independent internal	PatientID_0110	PatientID_0110_T1_to_T2_t1c	60	Male	GBM	4	80
Independent internal	PatientID_0114	PatientID_0114_T1_to_T2_t1c	77	Male	GBM	4	8
Independent internal	PatientID_0116	PatientID_0116_T1_to_T2_t1c	81	Male	GBM	4	49
Independent internal	PatientID_0117	PatientID_0117_T1_to_T2_t1c	68	Male	GBM	4	69
Independent internal	PatientID_0118	PatientID_0118_T1_to_T2_t1c	66	Male	GBM	4	90
Independent internal	PatientID_0121	PatientID_0121_T1_to_T2_t1c	52	Female	GBM	4	91
Independent internal	PatientID_0123	PatientID_0123_T1_to_T2_t1c	69	Male	GBM	4	12
Independent internal	PatientID_0125	PatientID_0125_T1_to_T2_t1c	76	Male	GBM	4	18
Independent internal	PatientID_0135	PatientID_0135_T1_to_T2_t1c	64	Male	GBM	4	53

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Independent internal	PatientID_0136	PatientID_0136_T1_to_T2_t1c	51	Female	GBM	4	54
Independent internal	PatientID_0138	PatientID_0138_T1_to_T2_t1c	65	Male	GBM	4	68
Independent internal	PatientID_0140	PatientID_0140_T1_to_T2_t1c	74	Male	GBM	4	109
Independent internal	PatientID_0142	PatientID_0142_T1_to_T2_t1c	72	Male	GBM	4	109
Independent internal	PatientID_0145	PatientID_0145_T1_to_T2_t1c	70	Male	GBM	4	140
Independent internal	PatientID_0147	PatientID_0147_T1_to_T2_t1c	71	Female	GBM	4	50
Independent internal	PatientID_0148	PatientID_0148_T1_to_T2_t1c	61	Female	GBM	4	10
Independent internal	PatientID_0149	PatientID_0149_T1_to_T2_t1c	47	Female	GBM	4	45
Independent internal	PatientID_0150	PatientID_0150_T1_to_T2_t1c	59	Female	GBM	4	91
Independent internal	PatientID_0151	PatientID_0151_T1_to_T2_t1c	70	Male	GBM	4	59
Independent internal	PatientID_0152	PatientID_0152_T1_to_T2_t1c	59	Female	GBM	4	46
Independent internal	PatientID_0156	PatientID_0156_T1_to_T2_t1c	66	Female	GBM	4	56
Independent internal	PatientID_0157	PatientID_0157_T1_to_T2_t1c	62	Female	GBM	4	53
Independent internal	PatientID_0159	PatientID_0159_T1_to_T2_t1c	85	Male	GBM	4	105
Independent internal	PatientID_0160	PatientID_0160_T1_to_T2_t1c	69	Male	GBM	4	50
Independent internal	PatientID_0162	PatientID_0162_T1_to_T2_t1c	66	Male	GBM	4	62
Independent internal	PatientID_0164	PatientID_0164_T1_to_T2_t1c	78	Male	GBM	4	21
Independent internal	PatientID_0169	PatientID_0169_T1_to_T2_t1c	77	Male	GBM	4	28
Independent internal	PatientID_0170	PatientID_0170_T1_to_T2_t1c	61	Male	GBM	4	91
Independent internal	PatientID_0171	PatientID_0171_T1_to_T2_t1c	68	Female	GBM	4	34
Independent internal	PatientID_0173	PatientID_0173_T1_to_T2_t1c	38	Female	GBM	4	25

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Independent internal	PatientID_0186	PatientID_0186_T1_to_T2_t1c	39	Female	Astrocytoma	2	36
Independent internal	PatientID_0187	PatientID_0187_T1_to_T2_t1c	48	Male	Astrocytoma	2	77
Independent internal	PatientID_0188	PatientID_0188_T1_to_T2_t1c	63	Female	Astrocytoma	3	26
Independent internal	PatientID_0189	PatientID_0189_T1_to_T2_t1c	64	Male	Pilocytic astrocytoma	1	182
Independent internal	PatientID_0191	PatientID_0191_T2_to_T3_t1c	34	Male	Astrocytoma	3	83
Independent internal	PatientID_0192	PatientID_0192_T1_to_T2_t1c	53	Female	Astrocytoma	2	208
Independent internal	PatientID_0195	PatientID_0195_T1_to_T2_t1c	38	Female	Astrocytoma	2	139
Independent internal	PatientID_0196	PatientID_0196_T1_to_T2_t1c	30	Male	Astrocytoma	2	103
Independent internal	PatientID_0197	PatientID_0197_T1_to_T2_t1c	31	Female	Astrocytoma	2	56
Independent internal	PatientID_0198	PatientID_0198_T1_to_T2_t1c	31	Female	Diffuse glioma	4	22
Independent internal	PatientID_0199	PatientID_0199_T1_to_T2_t1c	23	Female	Diffuse glioma	4	29
Independent internal	PatientID_0201	PatientID_0201_T1_to_T2_t1c	45	Male	Diffuse glioma	4	42
Independent internal	PatientID_0202	PatientID_0202_T1_to_T2_t1c	32	Male	Astrocytoma	2	22
Independent internal	PatientID_0204	PatientID_0204_T1_to_T2_t1c	24	Female	Diffuse glioma	4	77
Independent internal	PatientID_0205	PatientID_0205_T1_to_T2_t1c	24	Female	Astrocytoma	2	23
Independent internal	PatientID_0207	PatientID_0207_T1_to_T2_t1c	35	Male	Diffuse glioma	4	142
Independent internal	PatientID_0208	PatientID_0208_T1_to_T2_t1c	20	Male	Pilocytic astrocytoma	1	371
Independent internal	PatientID_0209	PatientID_0209_T1_to_T2_t1c	45	Male	Diffuse glioma	4	97
Independent internal	PatientID_0210	PatientID_0210_T1_to_T2_t1c	26	Female	Astrocytoma	2	24
Independent internal	PatientID_0211	PatientID_0211_T1_to_T2_t1c	79	Female	Astrocytoma	2	13
Independent internal	PatientID_0212	PatientID_0212_T1_to_T2_t1c	72	Male	Astrocytoma	2	54

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Independent internal	PatientID_0213	PatientID_0213_T1_to_T2_t1c	31	Male	Astrocytoma	2	3
Independent internal	PatientID_0214	PatientID_0214_T1_to_T2_t1c	30	Male	Astrocytoma	2	96
Independent internal	PatientID_0215	PatientID_0215_T1_to_T2_t1c	29	Female	Diffuse glioma	4	56
Independent internal	PatientID_0220	PatientID_0220_T1_to_T6_t1c	33	Female	Astrocytoma	2	
Independent internal	PatientID_0235	PatientID_0235_T1_to_T3_t1c	32	Female	Astrocytoma	2	183
Independent internal	PatientID_0236	PatientID_0236_T1_to_T3_t1c	25	Male	Astrocytoma	2	126
Independent internal	PatientID_0237	PatientID_0237_T1_to_T3_t1c	71	Male	Diffuse glioma	4	11
Independent internal	PatientID_0238	PatientID_0238_T1_to_T3_t1c	54	Female	GBM	4	92
Independent internal	PatientID_0239	PatientID_0239_T1_to_T3_t1c	78	Female	GBM	4	183
Independent internal	PatientID_0240	PatientID_0240_T1_to_T3_t1c	72	Male	Astrocytoma	2	182
Independent internal	PatientID_0242	PatientID_0242_T1_to_T3_t1c	28	Female	Oligodendro-glioma	2	245
Independent internal	PatientID_0249	PatientID_0249_T1_to_T3_t1c	48	Male	Astrocytoma	2	182
Independent internal	PatientID_0250	PatientID_0250_T1_to_T2_t1c	59	Female	GBM	4	70
Independent internal	PatientID_0251	PatientID_0251_T1_to_T2_t1c	78	Female	GBM	4	77
Independent internal	PatientID_0252	PatientID_0252_T1_to_T2_t1c	63	Male	GBM	4	43
Independent internal	PatientID_0254	PatientID_0254_T1_to_T2_t1c	46	Female	GBM	4	68
Independent internal	PatientID_0255	PatientID_0255_T1_to_T2_t1c	62	Female	Glioma w/ GBM features	4	56
Independent internal	PatientID_0256	PatientID_0256_T1_to_T2_t1c	31	Female	Diffuse glioma	4	22
Independent internal	PatientID_0259	PatientID_0259_T1_to_T2_t1c	43	Male	Astrocytoma	2	133
Independent internal	PatientID_0260	PatientID_0260_T1_to_T2_t1c	69	Female	GBM	4	84
Independent internal	PatientID_0264	PatientID_0264_T1_to_T2_t1c	36	Male	Astrocytoma	2	115

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
Independent internal	PatientID_0265	PatientID_0265_T1_to_T2_t1c	52	Male	Oligodendro-glioma	2	371
Independent internal	PatientID_0267	PatientID_0267_T1_to_T2_t1c	66	Male	GBM	4	62
Independent internal	PatientID_0268	PatientID_0268_T1_to_T3_t1c	62	Male	GBM	4	40
Independent internal	PatientID_0270	PatientID_0270_T1_to_T3_t1c	28	Male	Oligodendro-glioma	2	196
Independent internal	PatientID_0271	PatientID_0271_T1_to_T3_t1c	30	Female	Astrocytoma	2	150
Independent internal	PatientID_0272	PatientID_0272_T1_to_T3_t1c	56	Female	GBM	4	119
Independent internal	PatientID_0274	PatientID_0274_T1_to_T3_t1c	61	Male	GBM	4	145
Independent internal	PatientID_0275	PatientID_0275_T1_to_T3_t1c	43	Female	Oligodendro-glioma	2	132
RHUH external	RHUH-0001	RHUH-0001_EARLY_POS TOP_TO_RECURRENCE _T1CE	65	Male	glioblastoma	4	
RHUH external	RHUH-0002	RHUH-0002_EARLY_POS TOP_TO_RECURRENCE _T1CE	65	Female	glioblastoma	4	
RHUH external	RHUH-0003	RHUH-0003_EARLY_POS TOP_TO_RECURRENCE _T1CE	75	Male	glioblastoma	4	
RHUH external	RHUH-0004	RHUH-0004_EARLY_POS TOP_TO_RECURRENCE _T1CE	64	Female	glioblastoma	4	
RHUH external	RHUH-0005	RHUH-0005_EARLY_POS TOP_TO_RECURRENCE _T1CE	52	Male	glioblastoma	4	
RHUH external	RHUH-0006	RHUH-0006_EARLY_POS TOP_TO_RECURRENCE _T1CE	72	Male	glioblastoma	4	
RHUH external	RHUH-0007	RHUH-0007_EARLY_POS TOP_TO_RECURRENCE _T1CE	51	Female	glioblastoma	4	
RHUH external	RHUH-0009	RHUH-0009_EARLY_POS TOP_TO_RECURRENCE _T1CE	68	Male	glioblastoma	4	
RHUH external	RHUH-0010	RHUH-0010_EARLY_POS TOP_TO_RECURRENCE _T1CE	66	Male	glioblastoma	4	

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
RHUH external	RHUH-0011	RHUH-0011_EARLY_POS TOP_TO_RECURRENCE _T1CE	71	Female	glioblastoma	4	
RHUH external	RHUH-0012	RHUH-0012_EARLY_POS TOP_TO_RECURRENCE _T1CE	64	Female	glioblastoma	4	
RHUH external	RHUH-0013	RHUH-0013_EARLY_POS TOP_TO_RECURRENCE _T1CE	54	Male	glioblastoma	4	
RHUH external	RHUH-0014	RHUH-0014_EARLY_POS TOP_TO_RECURRENCE _T1CE	78	Male	glioblastoma	4	
RHUH external	RHUH-0015	RHUH-0015_EARLY_POS TOP_TO_RECURRENCE _T1CE	60	Male	glioblastoma	4	
RHUH external	RHUH-0016	RHUH-0016_EARLY_POS TOP_TO_RECURRENCE _T1CE	68	Female	glioblastoma	4	
RHUH external	RHUH-0017	RHUH-0017_EARLY_POS TOP_TO_RECURRENCE _T1CE	47	Male	glioblastoma	4	
RHUH external	RHUH-0018	RHUH-0018_EARLY_POS TOP_TO_RECURRENCE _T1CE	57	Male	glioblastoma	4	
RHUH external	RHUH-0019	RHUH-0019_EARLY_POS TOP_TO_RECURRENCE _T1CE	61	Male	glioblastoma	4	
RHUH external	RHUH-0020	RHUH-0020_EARLY_POS TOP_TO_RECURRENCE _T1CE	55	Male	glioblastoma	4	
RHUH external	RHUH-0021	RHUH-0021_EARLY_POS TOP_TO_RECURRENCE _T1CE	56	Male	glioblastoma	4	
RHUH external	RHUH-0022	RHUH-0022_EARLY_POS TOP_TO_RECURRENCE _T1CE	78	Female	glioblastoma	4	
RHUH external	RHUH-0023	RHUH-0023_EARLY_POS TOP_TO_RECURRENCE _T1CE	59	Male	glioblastoma	4	
RHUH external	RHUH-0024	RHUH-0024_EARLY_POS TOP_TO_RECURRENCE _T1CE	50	Male	glioblastoma	4	
RHUH external	RHUH-0025	RHUH-0025_EARLY_POS TOP_TO_RECURRENCE _T1CE	49	Male	glioblastoma	4	

cohort	patient id	case id	age years	sex at birth	primary diagnosis	grade	current to future interval days
RHUH external	RHUH-0026	RHUH-0026_EARLY_POS TOP_TO_RECURRENCE _T1CE	63	Female	glioblastoma	4	
RHUH external	RHUH-0027	RHUH-0027_EARLY_POS TOP_TO_RECURRENCE _T1CE	77	Female	glioblastoma	4	
RHUH external	RHUH-0028	RHUH-0028_EARLY_POS TOP_TO_RECURRENCE _T1CE	70	Male	glioblastoma	4	
RHUH external	RHUH-0029	RHUH-0029_EARLY_POS TOP_TO_RECURRENCE _T1CE	75	Male	glioblastoma	4	
RHUH external	RHUH-0030	RHUH-0030_EARLY_POS TOP_TO_RECURRENCE _T1CE	54	Male	glioblastoma	4	
RHUH external	RHUH-0031	RHUH-0031_EARLY_POS TOP_TO_RECURRENCE _T1CE	75	Female	glioblastoma	4	
RHUH external	RHUH-0032	RHUH-0032_EARLY_POS TOP_TO_RECURRENCE _T1CE	54	Male	glioblastoma	4	
RHUH external	RHUH-0033	RHUH-0033_EARLY_POS TOP_TO_RECURRENCE _T1CE	74	Male	glioblastoma	4	
RHUH external	RHUH-0034	RHUH-0034_EARLY_POS TOP_TO_RECURRENCE _T1CE	66	Male	glioblastoma	4	
RHUH external	RHUH-0035	RHUH-0035_EARLY_POS TOP_TO_RECURRENCE _T1CE	55	Male	glioblastoma	4	
RHUH external	RHUH-0036	RHUH-0036_EARLY_POS TOP_TO_RECURRENCE _T1CE	60	Female	glioblastoma	4	
RHUH external	RHUH-0037	RHUH-0037_EARLY_POS TOP_TO_RECURRENCE _T1CE	45	Male	glioblastoma	4	
RHUH external	RHUH-0038	RHUH-0038_EARLY_POS TOP_TO_RECURRENCE _T1CE	69	Male	glioblastoma	4	
RHUH external	RHUH-0039	RHUH-0039_EARLY_POS TOP_TO_RECURRENCE _T1CE	67	Female	glioblastoma	4	
RHUH external	RHUH-0040	RHUH-0040_EARLY_POS TOP_TO_RECURRENCE _T1CE	60	Male	glioblastoma	4	

Part 2 of 2

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interval status	metadata source sha256
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AVAILABLE	4341badf954511d67fc52b82b3b488ed7323f7e46d249a354ff9b3a62269c79b
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AVAILABLE	4341badf954511d67fc52b82b3b488ed7323f7e46d249a354ff9b3a62269c79b
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AVAILABLE	4341badf954511d67fc52b82b3b488ed7323f7e46d249a354ff9b3a62269c79b
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Table S3. Internal secondary metrics

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method	metric	n	mean	SD	median	Q1	Q3
Fixed	IoU_0.5	113	0.1428	0.09134	0.1363	0.08025	0.1872
Fixed	precision_0.5	113	0.2647	0.2096	0.2053	0.105	0.3947
Fixed	recall_0.5	113	0.4287	0.2542	0.3867	0.223	0.59
Fixed	soft_Dice	113	0.1852	0.112	0.1837	0.103	0.2431
Fixed	Brier	113	0.004523	0.004734	0.003303	0.001627	0.005121
Fixed	average_precision	113	0.2272	0.1624	0.188	0.09859	0.3366
Fixed	predicted_positive_volume	113	3.181e+04	2.191e+04	2.517e+04	1.541e+04	4.355e+04
Fixed	target_to_predicted_volume_ratio	113	1.25	1.837	0.5285	0.195	1.23
Naive	IoU_0.5	113	0.1428	0.09134	0.1363	0.08025	0.1872
Naive	precision_0.5	113	0.2647	0.2096	0.2053	0.105	0.3947
Naive	recall_0.5	113	0.4287	0.2542	0.3867	0.223	0.59
Naive	soft_Dice	113	0.2244	0.1269	0.2243	0.1395	0.2905
Naive	Brier	113	0.00478	0.004989	0.003434	0.00163	0.005516
Naive	average_precision	113	0.2267	0.1625	0.1878	0.09874	0.3366
Naive	predicted_positive_volume	113	3.181e+04	2.191e+04	2.517e+04	1.541e+04	4.355e+04
Naive	target_to_predicted_volume_ratio	113	1.25	1.837	0.5285	0.195	1.23
EIA-linear	IoU_0.5	113	0.1834	0.1103	0.1764	0.1088	0.2388
EIA-linear	precision_0.5	113	0.3132	0.2349	0.2503	0.1274	0.4548
EIA-linear	recall_0.5	113	0.508	0.2561	0.4872	0.3042	0.6902
EIA-linear	soft_Dice	113	0.2602	0.1492	0.2606	0.1485	0.385
EIA-linear	Brier	113	0.003341	0.002776	0.002608	0.001387	0.004138
EIA-linear	average_precision	113	0.3611	0.2364	0.3232	0.1654	0.5269
EIA-linear	predicted_positive_volume	113	3.223e+04	2.276e+04	2.592e+04	1.464e+04	4.6e+04
EIA-linear	target_to_predicted_volume_ratio	113	1.145	1.686	0.5302	0.2063	1.19
EIA-blend-0.90	IoU_0.5	113	0.1568	0.09854	0.1464	0.09721	0.2065
EIA-blend-0.90	precision_0.5	113	0.294	0.224	0.2377	0.1182	0.435
EIA-blend-0.90	recall_0.5	113	0.4342	0.2537	0.3848	0.2277	0.5965
EIA-blend-0.90	soft_Dice	113	0.2123	0.1212	0.2088	0.1241	0.2862

method	metric	n	mean	SD	median	Q1	Q3
EIA-blend-0.90	Brier	113	0.003808	0.003966	0.002786	0.001385	0.004373
EIA-blend-0.90	average_precision	113	0.3588	0.1943	0.3191	0.2157	0.5073
EIA-blend-0.90	predicted_positive_volume	113	2.892e+04	2.039e+04	2.255e+04	1.345e+04	4.069e+04
EIA-blend-0.90	target_to_predicted_volume_ratio	113	1.338	1.921	0.5724	0.2242	1.389
EIA-blend-0.75	IoU_0.5	113	0.1991	0.1174	0.1869	0.1311	0.263
EIA-blend-0.75	precision_0.5	113	0.3793	0.257	0.3351	0.1741	0.5898
EIA-blend-0.75	recall_0.5	113	0.4491	0.2492	0.4115	0.235	0.6172
EIA-blend-0.75	soft_Dice	113	0.2532	0.1413	0.2459	0.1475	0.3632
EIA-blend-0.75	Brier	113	0.002876	0.002962	0.002036	0.0009949	0.003301
EIA-blend-0.75	average_precision	113	0.4779	0.2157	0.4498	0.3277	0.6499
EIA-blend-0.75	predicted_positive_volume	113	2.305e+04	1.731e+04	1.924e+04	9285	3.193e+04
EIA-blend-0.75	target_to_predicted_volume_ratio	113	1.58	2.251	0.7209	0.295	1.957
EIA-morph	IoU_0.5	113	0.1511	0.09465	0.1358	0.09493	0.1974
EIA-morph	precision_0.5	113	0.2821	0.2184	0.2141	0.1117	0.4273
EIA-morph	recall_0.5	113	0.4297	0.2557	0.3844	0.2227	0.6041
EIA-morph	soft_Dice	113	0.2514	0.1372	0.2391	0.1734	0.3297
EIA-morph	Brier	113	0.005195	0.005021	0.003417	0.001792	0.006601
EIA-morph	average_precision	113	0.1059	0.08344	0.08012	0.04958	0.1501
EIA-morph	predicted_positive_volume	113	2.937e+04	2.184e+04	2.359e+04	1.28e+04	3.983e+04
EIA-morph	target_to_predicted_volume_ratio	113	3.795	26.86	0.5334	0.2071	1.328
Full PCC	IoU_0.5	113	0.3027	0.1498	0.3067	0.2104	0.3918
Full PCC	precision_0.5	113	0.499	0.2743	0.4785	0.2805	0.7087
Full PCC	recall_0.5	113	0.5404	0.215	0.5486	0.3767	0.7067
Full PCC	soft_Dice	113	0.2667	0.1496	0.2635	0.1626	0.3671
Full PCC	Brier	113	0.003102	0.004117	0.00172	0.0008422	0.003231
Full PCC	average_precision	113	0.4142	0.2117	0.4166	0.2552	0.5815
Full PCC	predicted_positive_volume	113	2.233e+04	1.85e+04	1.853e+04	8041	3.158e+04
Full PCC	target_to_predicted_volume_ratio	113	1.44	2.322	0.8266	0.4321	1.62
No-smoothing PCC	IoU_0.5	113	0.4958	0.2014	0.5164	0.363	0.629
No-smoothing PCC	precision_0.5	113	0.7088	0.2561	0.7618	0.5467	0.9502
No-smoothing PCC	recall_0.5	113	0.7034	0.244	0.7441	0.5275	0.9294

method	metric	n	mean	SD	median	Q1	Q3
No-smoothing PCC	soft_Dice	113	0.3204	0.1647	0.3259	0.2144	0.4415
No-smoothing PCC	Brier	113	0.002583	0.004011	0.001206	0.0005939	0.002499
No-smoothing PCC	average_precision	113	0.6473	0.1988	0.6639	0.5326	0.8112
No-smoothing PCC	predicted_positive_volume	113	2.03e+04	1.91e+04	1.492e+04	6355	2.606e+04
No-smoothing PCC	target_to_predicted_volume_ratio	113	1.514	2.212	1.034	0.7034	1.604

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IQR	bootstrap 95ci low	bootstrap 95ci high	bootstrap resamples	bootstrap seed
0.1069	0.1265	0.1601	10000	20260803
0.2897	0.2269	0.3036	10000	20260803
0.367	0.3814	0.4749	10000	20260803
0.1401	0.165	0.2062	10000	20260803
0.003494	0.003719	0.00544	10000	20260803
0.238	0.1977	0.2581	10000	20260803
2.814e+04	2.784e+04	3.585e+04	10000	20260803
1.035	0.9356	1.606	10000	20260803
0.1069	0.1264	0.1602	10000	20260803
0.2897	0.2275	0.3034	10000	20260803
0.367	0.3818	0.4759	10000	20260803
0.151	0.2013	0.2482	10000	20260803
0.003886	0.00392	0.005746	10000	20260803
0.2378	0.1975	0.2573	10000	20260803
2.814e+04	2.791e+04	3.593e+04	10000	20260803
1.035	0.9292	1.613	10000	20260803
0.13	0.1637	0.2037	10000	20260803
0.3274	0.2706	0.3565	10000	20260803
0.386	0.46	0.5541	10000	20260803
0.2365	0.2329	0.2883	10000	20260803
0.002751	0.002857	0.003871	10000	20260803
0.3614	0.3181	0.405	10000	20260803
3.136e+04	2.825e+04	3.651e+04	10000	20260803
0.9835	0.8602	1.471	10000	20260803
0.1092	0.1392	0.1756	10000	20260803
0.3167	0.253	0.3365	10000	20260803
0.3688	0.3878	0.4808	10000	20260803
0.1621	0.1901	0.2343	10000	20260803
0.002988	0.003123	0.004585	10000	20260803
0.2916	0.3237	0.3949	10000	20260803
2.724e+04	2.528e+04	3.273e+04	10000	20260803
1.165	1.012	1.709	10000	20260803

IQR	bootstrap 95ci low	bootstrap 95ci high	bootstrap resamples	bootstrap seed
0.1319	0.1773	0.2204	10000	20260803
0.4156	0.3327	0.4267	10000	20260803
0.3822	0.4037	0.4956	10000	20260803
0.2157	0.2274	0.2793	10000	20260803
0.002306	0.002359	0.003443	10000	20260803
0.3222	0.438	0.5179	10000	20260803
2.264e+04	1.996e+04	2.637e+04	10000	20260803
1.662	1.195	2.017	10000	20260803
0.1024	0.1343	0.1688	10000	20260803
0.3157	0.2432	0.3233	10000	20260803
0.3814	0.3835	0.4769	10000	20260803
0.1563	0.2259	0.2765	10000	20260803
0.004809	0.004324	0.006143	10000	20260803
0.1005	0.09089	0.1211	10000	20260803
2.703e+04	2.544e+04	3.348e+04	10000	20260803
1.121	1.024	9.061	10000	20260803
0.1814	0.2753	0.3302	10000	20260803
0.4282	0.4483	0.5492	10000	20260803
0.33	0.5006	0.5803	10000	20260803
0.2045	0.2396	0.2943	10000	20260803
0.002389	0.002414	0.003921	10000	20260803
0.3264	0.3757	0.4536	10000	20260803
2.354e+04	1.904e+04	2.574e+04	10000	20260803
1.188	1.072	1.904	10000	20260803
0.266	0.4589	0.533	10000	20260803
0.4035	0.6607	0.754	10000	20260803
0.4019	0.6575	0.7463	10000	20260803
0.2272	0.2905	0.3506	10000	20260803
0.001905	0.001913	0.003388	10000	20260803
0.2787	0.6107	0.6825	10000	20260803
1.97e+04	1.7e+04	2.382e+04	10000	20260803
0.9008	1.17	1.964	10000	20260803

Table S4. RHUH secondary metrics

Part 1 of 2

method	metric	n	mean	SD	median	Q1	Q3
Fixed	Dice_0.5	39	0.1895	0.1318	0.1811	0.08101	0.2697
Fixed	IoU_0.5	39	0.1105	0.08301	0.09956	0.04222	0.1559
Fixed	precision_0.5	39	0.2422	0.1968	0.2022	0.064	0.3866
Fixed	recall_0.5	39	0.2147	0.152	0.1606	0.1199	0.3086
Fixed	soft_Dice	39	0.1512	0.1064	0.1456	0.06035	0.2269
Fixed	Brier	39	0.009284	0.004699	0.007635	0.005454	0.01239
Fixed	average_precision	39	0.1531	0.1213	0.1417	0.04085	0.2387
Fixed	predicted_positive_volume	39	5.279e+04	2.36e+04	5.117e+04	3.662e+04	6.931e+04
Fixed	target_to_predicted_volume_ratio	39	1.494	1.55	0.933	0.6321	2.082
Naive	Dice_0.5	39	0.1895	0.1318	0.1811	0.08101	0.2697
Naive	IoU_0.5	39	0.1105	0.08301	0.09956	0.04222	0.1559
Naive	precision_0.5	39	0.2422	0.1968	0.2022	0.064	0.3866
Naive	recall_0.5	39	0.2147	0.152	0.1606	0.1199	0.3086
Naive	soft_Dice	39	0.179	0.1247	0.1616	0.07547	0.25
Naive	Brier	39	0.009386	0.005005	0.007521	0.005225	0.01299
Naive	average_precision	39	0.1528	0.1212	0.1396	0.04003	0.2387
Naive	predicted_positive_volume	39	5.279e+04	2.36e+04	5.117e+04	3.662e+04	6.931e+04
Naive	target_to_predicted_volume_ratio	39	1.494	1.55	0.933	0.6321	2.082
EIA-linear	Dice_0.5	39	0.2461	0.153	0.2291	0.1285	0.3463
EIA-linear	IoU_0.5	39	0.1489	0.1014	0.1293	0.06864	0.2094
EIA-linear	precision_0.5	39	0.3248	0.2462	0.2727	0.1358	0.5009
EIA-linear	recall_0.5	39	0.2612	0.1682	0.2373	0.1646	0.3857
EIA-linear	soft_Dice	39	0.2761	0.1357	0.2434	0.1891	0.4028
EIA-linear	Brier	39	0.006255	0.002851	0.005547	0.003857	0.008322
EIA-linear	average_precision	39	0.3541	0.2166	0.2854	0.1898	0.5348
EIA-linear	predicted_positive_volume	39	4.744e+04	2.609e+04	4.014e+04	2.888e+04	6.252e+04
EIA-linear	target_to_predicted_volume_ratio	39	1.716	1.755	1.007	0.7257	2.216
EIA-blend-0.90	Dice_0.5	39	0.2059	0.1393	0.2099	0.09195	0.2979

method	metric	n	mean	SD	median	Q1	Q3
EIA-blend-0.90	IoU_0.5	39	0.1215	0.08907	0.1172	0.0482	0.175
EIA-blend-0.90	precision_0.5	39	0.2761	0.219	0.2347	0.07691	0.4485
EIA-blend-0.90	recall_0.5	39	0.2211	0.1542	0.1721	0.1267	0.3218
EIA-blend-0.90	soft_Dice	39	0.1945	0.1145	0.1991	0.1066	0.2801
EIA-blend-0.90	Brier	39	0.007768	0.003957	0.006328	0.00455	0.01038
EIA-blend-0.90	average_precision	39	0.2722	0.1595	0.2916	0.1095	0.4156
EIA-blend-0.90	predicted_positive_volume	39	4.735e+04	2.248e+04	4.414e+04	3.199e+04	6.195e+04
EIA-blend-0.90	target_to_predicted_volume_ratio	39	1.667	1.697	1.056	0.7412	2.428
EIA-blend-0.75	Dice_0.5	39	0.2482	0.1566	0.2459	0.1243	0.3646
EIA-blend-0.75	IoU_0.5	39	0.1507	0.1039	0.1402	0.06627	0.2229
EIA-blend-0.75	precision_0.5	39	0.3631	0.2632	0.3176	0.138	0.6226
EIA-blend-0.75	recall_0.5	39	0.2394	0.1592	0.2103	0.1392	0.3574
EIA-blend-0.75	soft_Dice	39	0.2614	0.1309	0.238	0.1709	0.3832
EIA-blend-0.75	Brier	39	0.005783	0.002987	0.004744	0.003362	0.00774
EIA-blend-0.75	average_precision	39	0.4066	0.2094	0.4108	0.238	0.5872
EIA-blend-0.75	predicted_positive_volume	39	3.82e+04	2.093e+04	3.481e+04	2.26e+04	4.934e+04
EIA-blend-0.75	target_to_predicted_volume_ratio	39	2.095	2.031	1.305	0.9592	2.498
Full PCC	Dice_0.5	39	0.3654	0.1877	0.3913	0.2291	0.5175
Full PCC	IoU_0.5	39	0.2389	0.1379	0.2433	0.1295	0.3491
Full PCC	precision_0.5	39	0.5292	0.2845	0.5293	0.3425	0.7772
Full PCC	recall_0.5	39	0.3248	0.1759	0.3144	0.2341	0.4436
Full PCC	soft_Dice	39	0.2382	0.1445	0.2284	0.134	0.3585
Full PCC	Brier	39	0.006457	0.004132	0.005062	0.003083	0.008279
Full PCC	average_precision	39	0.3093	0.1903	0.3213	0.1503	0.4898
Full PCC	predicted_positive_volume	39	3.556e+04	2.366e+04	2.786e+04	1.898e+04	4.964e+04
Full PCC	target_to_predicted_volume_ratio	39	2.345	2.445	1.58	1.142	2.597
No-smoothing PCC	Dice_0.5	39	0.4512	0.2103	0.4658	0.3502	0.6201
No-smoothing PCC	IoU_0.5	39	0.3136	0.1698	0.3036	0.2125	0.4494
No-smoothing PCC	precision_0.5	39	0.6633	0.2889	0.7185	0.5209	0.8969
No-smoothing PCC	recall_0.5	39	0.3898	0.2012	0.4051	0.2868	0.5306
No-smoothing PCC	soft_Dice	39	0.2696	0.1557	0.2777	0.1653	0.4003

method	metric	n	mean	SD	median	Q1	Q3
No-smoothing PCC	Brier	39	0.005765	0.003959	0.004686	0.002675	0.007503
No-smoothing PCC	average_precision	39	0.4044	0.2091	0.4197	0.2725	0.5755
No-smoothing PCC	predicted_positive_volume	39	3.36e+04	2.391e+04	2.775e+04	1.596e+04	4.568e+04
No-smoothing PCC	target_to_predicted_volume_ratio	39	2.559	2.713	1.679	1.281	2.71

IQR	bootstrap 95ci low	bootstrap 95ci high	bootstrap replicates	bootstrap seed	secondary pairwise inference
0.1887	0.149	0.2306	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1137	0.08506	0.1365	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3226	0.1816	0.3029	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1887	0.1687	0.264	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1665	0.1182	0.1838	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.006939	0.007854	0.01076	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1978	0.1162	0.191	10000	20260810	NOT_PRELOCKED_NOT_RUN
3.269e+04	4.571e+04	6.026e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.45	1.066	2.015	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1887	0.149	0.2306	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1137	0.08506	0.1365	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3226	0.1816	0.3029	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1887	0.1687	0.264	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1746	0.1405	0.2176	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.007764	0.00786	0.01096	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1987	0.116	0.1908	10000	20260810	NOT_PRELOCKED_NOT_RUN
3.269e+04	4.571e+04	6.026e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.45	1.066	2.015	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2179	0.1983	0.2932	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1408	0.1175	0.1804	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3651	0.2492	0.4024	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2211	0.2102	0.315	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2137	0.2337	0.317	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.004465	0.005375	0.007154	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.345	0.2868	0.4218	10000	20260810	NOT_PRELOCKED_NOT_RUN
3.364e+04	3.965e+04	5.563e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.49	1.224	2.295	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.206	0.1628	0.2494	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1268	0.09425	0.1493	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3715	0.2087	0.3442	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1951	0.1744	0.2709	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1735	0.1593	0.2292	10000	20260810	NOT_PRELOCKED_NOT_RUN

IQR	bootstrap 95ci low	bootstrap 95ci high	bootstrap replicates	bootstrap seed	secondary pairwise inference
0.005826	0.006563	0.009014	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.306	0.2226	0.3215	10000	20260810	NOT_PRELOCKED_NOT_RUN
2.996e+04	4.065e+04	5.448e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.686	1.197	2.236	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2403	0.1995	0.2969	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.1567	0.1187	0.1829	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.4846	0.282	0.4447	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2181	0.191	0.2905	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2123	0.2204	0.3007	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.004378	0.004878	0.006726	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3491	0.3412	0.4701	10000	20260810	NOT_PRELOCKED_NOT_RUN
2.674e+04	3.194e+04	4.483e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.539	1.52	2.758	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2883	0.3064	0.4214	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2196	0.1959	0.2805	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.4346	0.4403	0.6149	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2095	0.2713	0.3802	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2245	0.1935	0.2817	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.005196	0.00523	0.007775	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3395	0.2502	0.3676	10000	20260810	NOT_PRELOCKED_NOT_RUN
3.066e+04	2.839e+04	4.294e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.454	1.687	3.175	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2699	0.3853	0.5142	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2368	0.2614	0.3653	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.376	0.5707	0.748	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.2438	0.3286	0.4524	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.235	0.2215	0.3165	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.004828	0.004597	0.007045	10000	20260810	NOT_PRELOCKED_NOT_RUN
0.3031	0.3396	0.4687	10000	20260810	NOT_PRELOCKED_NOT_RUN
2.972e+04	2.63e+04	4.108e+04	10000	20260810	NOT_PRELOCKED_NOT_RUN
1.429	1.835	3.492	10000	20260810	NOT_PRELOCKED_NOT_RUN

Table S5. Development mechanism and method summaries

Part 1 of 2

condition	metric	N	mean	SD	median	Q1	Q3
FACTORIAL_ERROR_OF_F_SUPPRESSION_OFF	dice	40	0.2762	0.1665	0.2682	0.1605	0.3941
FACTORIAL_ERROR_OF_F_SUPPRESSION_OFF	iou	40	0.1722	0.1283	0.1549	0.08725	0.2454
FACTORIAL_ERROR_OF_F_SUPPRESSION_ON	dice	40	0.2881	0.1661	0.2717	0.163	0.4126
FACTORIAL_ERROR_OF_F_SUPPRESSION_ON	iou	40	0.1805	0.1309	0.1572	0.08871	0.2599
FACTORIAL_ERROR_ON_SUPPRESSION_OFF	dice	40	0.3612	0.1784	0.3648	0.224	0.4674
FACTORIAL_ERROR_ON_SUPPRESSION_OFF	iou	40	0.2358	0.1468	0.2231	0.1261	0.305
FACTORIAL_ERROR_ON_SUPPRESSION_ON	dice	40	0.3884	0.1735	0.3775	0.2645	0.4937
FACTORIAL_ERROR_ON_SUPPRESSION_ON	iou	40	0.2566	0.1501	0.2326	0.1524	0.3278
FIXED_P0	dice	40	0.2762	0.1665	0.2682	0.1605	0.3941
FIXED_P0	iou	40	0.1722	0.1283	0.1549	0.08725	0.2454
FULL_PCC	dice	40	0.3884	0.1735	0.3775	0.2645	0.4937
FULL_PCC	iou	40	0.2566	0.1501	0.2326	0.1524	0.3278
GLOBAL_DISCREPANCY	dice	40	0.3932	0.1715	0.3786	0.2786	0.4983
GLOBAL_DISCREPANCY	iou	40	0.2601	0.1494	0.2335	0.1619	0.3318
NAIVE	dice	40	0.275	0.1674	0.2643	0.1546	0.3941
NAIVE	iou	40	0.1715	0.1288	0.1523	0.08376	0.2454
NO_ERROR_GUIDED_TERM	dice	40	0.2881	0.1661	0.2717	0.163	0.4126
NO_ERROR_GUIDED_TERM	iou	40	0.1805	0.1309	0.1572	0.08871	0.2599
NO_OUTSIDE_SUPPRESSION	dice	40	0.3612	0.1784	0.3648	0.224	0.4674
NO_OUTSIDE_SUPPRESSION	iou	40	0.2358	0.1468	0.2231	0.1261	0.305
NO_SMOOTHING	dice	40	0.5001	0.1764	0.5134	0.3788	0.5945
NO_SMOOTHING	iou	40	0.3529	0.1718	0.3453	0.2337	0.4231
ONE_ROUND	dice	40	0.2876	0.1675	0.2811	0.1655	0.4117

condition	metric	N	mean	SD	median	Q1	Q3
ONE_ROUND	iou	40	0.1803	0.1307	0.1636	0.09023	0.2592
FIXED_P0	dice	40	0.2762	0.1665	0.2682	0.1605	0.3941
FIXED_P0	iou	40	0.1722	0.1283	0.1549	0.08725	0.2454
FULL_PCC	dice	40	0.3884	0.1735	0.3775	0.2645	0.4937
FULL_PCC	iou	40	0.2566	0.1501	0.2326	0.1524	0.3278
NAIVE	dice	40	0.275	0.1674	0.2643	0.1546	0.3941
NAIVE	iou	40	0.1715	0.1288	0.1523	0.08376	0.2454
SHUFFLED_TARGET_PC C	dice	40	0.2836	0.1756	0.2625	0.1576	0.3935
SHUFFLED_TARGET_PC C	iou	40	0.1798	0.1477	0.1511	0.08554	0.2449
CLEAN	dice	40	0.5328	0.2267	0.57	0.3548	0.6961
CLEAN	iou	40	0.3954	0.2164	0.3986	0.2157	0.5339
CLEAN	dice	40	0.3499	0.2208	0.3008	0.1765	0.4478
CLEAN	iou	40	0.2369	0.192	0.177	0.09678	0.2885
CLEAN	dice	40	0.3884	0.1735	0.3775	0.2645	0.4937
CLEAN	iou	40	0.2566	0.1501	0.2326	0.1524	0.3278
FP_25	dice	40	0.5322	0.2267	0.5687	0.3541	0.6955
FP_25	iou	40	0.3948	0.2164	0.3974	0.2152	0.5333
FP_25	dice	40	0.3431	0.2221	0.299	0.1763	0.4358
FP_25	iou	40	0.2319	0.192	0.1758	0.09668	0.2786
FP_25	dice	40	0.3842	0.1739	0.3754	0.2585	0.4913
FP_25	iou	40	0.2533	0.1496	0.2311	0.1485	0.3257
MIXED	dice	40	0.4542	0.1954	0.458	0.3103	0.5976
MIXED	iou	40	0.3151	0.1716	0.2978	0.1837	0.4263
MIXED	dice	40	0.325	0.2025	0.2923	0.168	0.4254
MIXED	iou	40	0.2133	0.1646	0.1712	0.09173	0.2702
MIXED	dice	40	0.3328	0.1674	0.3337	0.1998	0.4546
MIXED	iou	40	0.2128	0.1361	0.2003	0.111	0.2942
PARTIAL_25	dice	40	0.4596	0.2018	0.4754	0.2844	0.6091
PARTIAL_25	iou	40	0.3205	0.1749	0.3119	0.1658	0.4379
PARTIAL_25	dice	40	0.3276	0.1977	0.2781	0.1698	0.4331
PARTIAL_25	iou	40	0.2144	0.1618	0.1615	0.09279	0.2764
PARTIAL_25	dice	40	0.3237	0.1641	0.315	0.2018	0.4399

condition	metric	N	mean	SD	median	Q1	Q3
PARTIAL_25	iou	40	0.2057	0.1331	0.1869	0.1123	0.282
PARTIAL_50	dice	40	0.4923	0.2147	0.5206	0.3299	0.6263
PARTIAL_50	iou	40	0.3536	0.197	0.3519	0.1975	0.456
PARTIAL_50	dice	40	0.3393	0.2114	0.2959	0.1724	0.4435
PARTIAL_50	iou	40	0.2262	0.1787	0.1737	0.09436	0.2849
PARTIAL_50	dice	40	0.3492	0.1705	0.343	0.2193	0.4664
PARTIAL_50	iou	40	0.2256	0.1417	0.207	0.1231	0.3041
SHIFT_3	dice	40	0.4845	0.1984	0.5021	0.3484	0.6273
SHIFT_3	iou	40	0.3428	0.1788	0.3354	0.2109	0.457
SHIFT_3	dice	40	0.3405	0.2077	0.2983	0.1722	0.4456
SHIFT_3	iou	40	0.2261	0.1725	0.1753	0.09422	0.2866
SHIFT_3	dice	40	0.3684	0.1713	0.363	0.241	0.4825
SHIFT_3	iou	40	0.2405	0.1442	0.2218	0.1374	0.318
BOUNDARY_TOLERANT	dice	40	0.3503	0.1792	0.3478	0.2154	0.4727
BOUNDARY_TOLERANT	iou	40	0.228	0.1497	0.2105	0.1207	0.3096
BOUNDARY_TOLERANT	dice	40	0.169	0.1604	0.1047	0.03317	0.3061
BOUNDARY_TOLERANT	iou	40	0.1011	0.1034	0.05527	0.01687	0.1807
ERODED_CURRENT	dice	40	0.4131	0.1643	0.3903	0.2989	0.5045
ERODED_CURRENT	iou	40	0.2749	0.1457	0.2424	0.1757	0.3373
ERODED_CURRENT	dice	40	0.3266	0.1419	0.3153	0.2117	0.408
ERODED_CURRENT	iou	40	0.204	0.1077	0.1873	0.1184	0.2563
LARGE_CHANGE_ONLY	dice	40	0.3884	0.1737	0.3776	0.2661	0.494
LARGE_CHANGE_ONLY	iou	40	0.2566	0.1505	0.2327	0.1534	0.3281
LARGE_CHANGE_ONLY	dice	40	0.3842	0.1781	0.3775	0.2609	0.4942
LARGE_CHANGE_ONLY	iou	40	0.2539	0.1524	0.2326	0.15	0.3282
ORIGINAL	dice	40	0.3884	0.1735	0.3775	0.2645	0.4937
ORIGINAL	iou	40	0.2566	0.1501	0.2326	0.1524	0.3278
ORIGINAL	dice	40	0.3884	0.1735	0.3775	0.2645	0.4937
ORIGINAL	iou	40	0.2566	0.1501	0.2326	0.1524	0.3278
FIXED_P0	dice	40	0.2762	0.1665	0.2682	0.1605	0.3941
FIXED_P0	iou	40	0.1722	0.1283	0.1549	0.08725	0.2454
RETROSPECTIVE_FUTURE_IMAGE_DIFFERENCE_CONTROL	dice	40	0.04745	0.05317	0.02651	0.002002	0.088

condition	metric	N	mean	SD	median	Q1	Q3
RETROSPECTIVE_FUTURE_IMAGE_DIFFERENCE_CONTROL	iou	40	0.02506	0.02869	0.01344	0.001002	0.04604

Part 2 of 2

bootstrap 95ci low	bootstrap 95ci high	family	method	evaluation mode
0.2265	0.3292	mechanism		
0.1354	0.2133	mechanism		
0.2392	0.3403	mechanism		
0.1432	0.2237	mechanism		
0.3087	0.4166	mechanism		
0.1925	0.2825	mechanism		
0.3364	0.4413	mechanism		
0.2131	0.3052	mechanism		
0.2272	0.3283	mechanism		
0.1349	0.2126	mechanism		
0.3367	0.4428	mechanism		
0.2131	0.3055	mechanism		
0.341	0.4477	mechanism		
0.2161	0.3092	mechanism		
0.2248	0.3281	mechanism		
0.1345	0.2143	mechanism		
0.2392	0.3405	mechanism		
0.1428	0.2229	mechanism		
0.3074	0.4169	mechanism		
0.1936	0.2835	mechanism		
0.4466	0.5545	mechanism		
0.3023	0.4071	mechanism		
0.2387	0.3406	mechanism		
0.1432	0.2215	mechanism		
0.2272	0.3283	counterfactual		
0.1349	0.2126	counterfactual		
0.3367	0.4428	counterfactual		
0.2131	0.3055	counterfactual		
0.2248	0.3281	counterfactual		
0.1345	0.2143	counterfactual		
0.2338	0.34	counterfactual		
0.1392	0.2289	counterfactual		

bootstrap 95ci low	bootstrap 95ci high	family	method	evaluation mode
0.4636	0.6005	imperfect_guidance	EIA_BLEND_075	
0.3288	0.4616	imperfect_guidance	EIA_BLEND_075	
0.2834	0.4187	imperfect_guidance	EIA_LINEAR	
0.1815	0.2982	imperfect_guidance	EIA_LINEAR	
0.3378	0.4412	imperfect_guidance	PCC	
0.213	0.305	imperfect_guidance	PCC	
0.4625	0.6011	imperfect_guidance	EIA_BLEND_075	
0.3297	0.4614	imperfect_guidance	EIA_BLEND_075	
0.2775	0.4141	imperfect_guidance	EIA_LINEAR	
0.1763	0.295	imperfect_guidance	EIA_LINEAR	
0.3319	0.4384	imperfect_guidance	PCC	
0.2098	0.3006	imperfect_guidance	PCC	
0.394	0.515	imperfect_guidance	EIA_BLEND_075	
0.2641	0.3692	imperfect_guidance	EIA_BLEND_075	
0.2645	0.388	imperfect_guidance	EIA_LINEAR	
0.1665	0.2669	imperfect_guidance	EIA_LINEAR	
0.2841	0.3857	imperfect_guidance	PCC	
0.1728	0.2569	imperfect_guidance	PCC	
0.3982	0.5202	imperfect_guidance	EIA_BLEND_075	
0.2679	0.3739	imperfect_guidance	EIA_BLEND_075	
0.2693	0.3905	imperfect_guidance	EIA_LINEAR	
0.1672	0.2652	imperfect_guidance	EIA_LINEAR	
0.2747	0.3761	imperfect_guidance	PCC	
0.1671	0.249	imperfect_guidance	PCC	
0.4264	0.5576	imperfect_guidance	EIA_BLEND_075	
0.2931	0.4149	imperfect_guidance	EIA_BLEND_075	
0.276	0.4056	imperfect_guidance	EIA_LINEAR	
0.1752	0.282	imperfect_guidance	EIA_LINEAR	
0.2988	0.4037	imperfect_guidance	PCC	
0.1847	0.2709	imperfect_guidance	PCC	
0.4247	0.5431	imperfect_guidance	EIA_BLEND_075	
0.2887	0.3995	imperfect_guidance	EIA_BLEND_075	
0.2793	0.4065	imperfect_guidance	EIA_LINEAR	

bootstrap 95ci low	bootstrap 95ci high	family	method	evaluation mode
0.175	0.2818	imperfect_guidance	EIA_LINEAR	
0.317	0.4229	imperfect_guidance	PCC	
0.1988	0.2853	imperfect_guidance	PCC	
0.2971	0.4057	target_construction		original_clean
0.1855	0.2762	target_construction		original_clean
0.121	0.2206	target_construction		same_definition
0.07022	0.1343	target_construction		same_definition
0.3636	0.4648	target_construction		original_clean
0.2329	0.3215	target_construction		original_clean
0.2846	0.3718	target_construction		same_definition
0.1719	0.2377	target_construction		same_definition
0.3356	0.4422	target_construction		original_clean
0.2119	0.3047	target_construction		original_clean
0.3309	0.4391	target_construction		same_definition
0.2105	0.3016	target_construction		same_definition
0.3359	0.4419	target_construction		original_clean
0.2126	0.3037	target_construction		original_clean
0.3362	0.4426	target_construction		same_definition
0.2128	0.3036	target_construction		same_definition
0.2272	0.3283	difference_map_control		
0.1349	0.2126	difference_map_control		
0.03206	0.06436	difference_map_control		
0.0167	0.03409	difference_map_control		

Table S6. Imperfect-guidance robustness

Part 1 of 1

condition	method	patients	mean oracle topk Dice	mean oracle topk IoU	mean retained true target fraction	added false positive volume	rows
CLEAN	EIA_BLEND_075	40	NA	NA	NA	NA	40
CLEAN	EIA_LINEAR	40	NA	NA	NA	NA	40
CLEAN	PCC	40	NA	NA	NA	NA	40
FP_25	EIA_BLEND_075	40	0.5322	0.3948	1	7578	40
FP_25	EIA_LINEAR	40	0.3431	0.2319	1	7578	40
FP_25	PCC	40	0.3842	0.2533	1	7578	40
MIXED	EIA_BLEND_075	40	0.4542	0.3151	0.5003	7578	40
MIXED	EIA_LINEAR	40	0.325	0.2133	0.5003	7578	40
MIXED	PCC	40	0.3328	0.2128	0.5003	7578	40
PARTIAL_25	EIA_BLEND_075	40	0.4596	0.3205	0.2505	0	40
PARTIAL_25	EIA_LINEAR	40	0.3276	0.2144	0.2505	0	40
PARTIAL_25	PCC	40	0.3237	0.2057	0.2505	0	40
PARTIAL_50	EIA_BLEND_075	40	0.4923	0.3536	0.5003	0	40
PARTIAL_50	EIA_LINEAR	40	0.3393	0.2262	0.5003	0	40
PARTIAL_50	PCC	40	0.3492	0.2256	0.5003	0	40
SHIFT_3	EIA_BLEND_075	40	0.4845	0.3428	0.5657	1.002e+04	40
SHIFT_3	EIA_LINEAR	40	0.3405	0.2261	0.5657	1.002e+04	40
SHIFT_3	PCC	40	0.3684	0.2405	0.5657	1.002e+04	40

Table S7. No-smoothing robustness

Part 1 of 2

condition	method	metric	N	mean	SD	median	Q1
CLEAN	FULL_PCC	dice_topk	40	0.3884	0.1735	0.3775	0.2645
CLEAN	FULL_PCC	iou_topk	40	0.2566	0.1501	0.2326	0.1524
CLEAN	FULL_PCC	dice_fixed05	40	0.2758	0.164	0.2688	0.1352
CLEAN	FULL_PCC	iou_fixed05	40	0.1714	0.1244	0.1553	0.07251
CLEAN	FULL_PCC	target_mass	40	1.368e+04	1.464e+04	8322	2986
CLEAN	FULL_PCC	false_guidance_mass	40	0	0	0	0
CLEAN	FULL_PCC	outside_clean_target_mas s	40	3.904e+04	2.388e+04	3.363e+04	2.222e+04
CLEAN	FULL_PCC	predicted_positive_voxels	40	4.977e+04	3.051e+04	4.272e+04	2.506e+04
CLEAN	NO_SMOOTHING_PCC	dice_topk	40	0.5001	0.1764	0.5134	0.3788
CLEAN	NO_SMOOTHING_PCC	iou_topk	40	0.3529	0.1718	0.3453	0.2337
CLEAN	NO_SMOOTHING_PCC	dice_fixed05	40	0.3263	0.1798	0.311	0.1814
CLEAN	NO_SMOOTHING_PCC	iou_fixed05	40	0.2097	0.1428	0.1841	0.0998
CLEAN	NO_SMOOTHING_PCC	target_mass	40	1.488e+04	1.524e+04	9762	3480
CLEAN	NO_SMOOTHING_PCC	false_guidance_mass	40	0	0	0	0
CLEAN	NO_SMOOTHING_PCC	outside_clean_target_mas s	40	3.481e+04	2.202e+04	3.027e+04	1.974e+04
CLEAN	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.548e+04	2.808e+04	3.832e+04	2.288e+04
FP_25	FULL_PCC	dice_topk	40	0.3842	0.1739	0.3754	0.2585
FP_25	FULL_PCC	iou_topk	40	0.2533	0.1496	0.2311	0.1485
FP_25	FULL_PCC	dice_fixed05	40	0.2728	0.1632	0.265	0.1319
FP_25	FULL_PCC	iou_fixed05	40	0.1692	0.1233	0.1528	0.07062
FP_25	FULL_PCC	target_mass	40	1.368e+04	1.464e+04	8322	2986
FP_25	FULL_PCC	false_guidance_mass	40	240.2	206.5	201.6	81.12
FP_25	FULL_PCC	outside_clean_target_mas s	40	3.972e+04	2.423e+04	3.392e+04	2.293e+04
FP_25	FULL_PCC	predicted_positive_voxels	40	5.07e+04	3.094e+04	4.331e+04	2.613e+04
FP_25	NO_SMOOTHING_PCC	dice_topk	40	0.4978	0.1759	0.511	0.3774
FP_25	NO_SMOOTHING_PCC	iou_topk	40	0.3506	0.1711	0.3432	0.2326
FP_25	NO_SMOOTHING_PCC	dice_fixed05	40	0.3252	0.1792	0.3104	0.1807
FP_25	NO_SMOOTHING_PCC	iou_fixed05	40	0.2089	0.1421	0.1838	0.09933

condition	method	metric	N	mean	SD	median	Q1
FP_25	NO_SMOOTHING_PCC	target_mass	40	1.488e+04	1.524e+04	9762	3480
FP_25	NO_SMOOTHING_PCC	false_guidance_mass	40	489.2	391.7	382.4	162.3
FP_25	NO_SMOOTHING_PCC	outside_clean_target_mas s	40	3.509e+04	2.207e+04	3.048e+04	2.007e+04
FP_25	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.573e+04	2.818e+04	3.873e+04	2.307e+04
MIXED	FULL_PCC	dice_topk	40	0.3328	0.1674	0.3337	0.1998
MIXED	FULL_PCC	iou_topk	40	0.2128	0.1361	0.2003	0.111
MIXED	FULL_PCC	dice_fixed05	40	0.2541	0.1592	0.2333	0.1185
MIXED	FULL_PCC	iou_fixed05	40	0.1561	0.1201	0.1321	0.06296
MIXED	FULL_PCC	target_mass	40	1.236e+04	1.37e+04	7228	2476
MIXED	FULL_PCC	false_guidance_mass	40	815	756.9	479	269.4
MIXED	FULL_PCC	outside_clean_target_mas s	40	3.928e+04	2.414e+04	3.384e+04	2.265e+04
MIXED	FULL_PCC	predicted_positive_voxels	40	4.928e+04	3.073e+04	4.181e+04	2.549e+04
MIXED	NO_SMOOTHING_PCC	dice_topk	40	0.3347	0.1703	0.3299	0.1956
MIXED	NO_SMOOTHING_PCC	iou_topk	40	0.2146	0.1365	0.1977	0.1085
MIXED	NO_SMOOTHING_PCC	dice_fixed05	40	0.2616	0.1646	0.2303	0.1244
MIXED	NO_SMOOTHING_PCC	iou_fixed05	40	0.1621	0.1265	0.1303	0.06632
MIXED	NO_SMOOTHING_PCC	target_mass	40	1.174e+04	1.314e+04	6844	2306
MIXED	NO_SMOOTHING_PCC	false_guidance_mass	40	1269	1075	807.5	452.9
MIXED	NO_SMOOTHING_PCC	outside_clean_target_mas s	40	3.543e+04	2.214e+04	3.056e+04	2.026e+04
MIXED	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.321e+04	2.824e+04	3.645e+04	2.155e+04
PARTIAL_25	FULL_PCC	dice_topk	40	0.3237	0.1641	0.315	0.2018
PARTIAL_25	FULL_PCC	iou_topk	40	0.2057	0.1331	0.1869	0.1123
PARTIAL_25	FULL_PCC	dice_fixed05	40	0.2505	0.1592	0.2199	0.1178
PARTIAL_25	FULL_PCC	iou_fixed05	40	0.1538	0.121	0.1235	0.06258
PARTIAL_25	FULL_PCC	target_mass	40	1.18e+04	1.327e+04	6934	2315
PARTIAL_25	FULL_PCC	false_guidance_mass	40	0	0	0	0
PARTIAL_25	FULL_PCC	outside_clean_target_mas s	40	3.822e+04	2.371e+04	3.321e+04	2.177e+04
PARTIAL_25	FULL_PCC	predicted_positive_voxels	40	4.754e+04	3.012e+04	4.042e+04	2.388e+04
PARTIAL_25	NO_SMOOTHING_PCC	dice_topk	40	0.3322	0.1591	0.3267	0.211
PARTIAL_25	NO_SMOOTHING_PCC	iou_topk	40	0.211	0.1277	0.1953	0.118
PARTIAL_25	NO_SMOOTHING_PCC	dice_fixed05	40	0.251	0.1627	0.2192	0.1206

condition	method	metric	N	mean	SD	median	Q1
PARTIAL_25	NO_SMOOTHING_PCC	iou_fixed05	40	0.1548	0.1252	0.1231	0.06415
PARTIAL_25	NO_SMOOTHING_PCC	target_mass	40	1.099e+04	1.252e+04	6490	2098
PARTIAL_25	NO_SMOOTHING_PCC	false_guidance_mass	40	0	0	0	0
PARTIAL_25	NO_SMOOTHING_PCC	outside_clean_target_mas s	40	3.481e+04	2.202e+04	3.027e+04	1.974e+04
PARTIAL_25	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.194e+04	2.799e+04	3.52e+04	2.043e+04
PARTIAL_50	FULL_PCC	dice_topk	40	0.3492	0.1705	0.343	0.2193
PARTIAL_50	FULL_PCC	iou_topk	40	0.2256	0.1417	0.207	0.1231
PARTIAL_50	FULL_PCC	dice_fixed05	40	0.2604	0.1612	0.2419	0.1235
PARTIAL_50	FULL_PCC	iou_fixed05	40	0.1606	0.1226	0.1376	0.06581
PARTIAL_50	FULL_PCC	target_mass	40	1.255e+04	1.383e+04	7327	2559
PARTIAL_50	FULL_PCC	false_guidance_mass	40	0	0	0	0
PARTIAL_50	FULL_PCC	outside_clean_target_mas s	40	3.845e+04	2.377e+04	3.333e+04	2.189e+04
PARTIAL_50	FULL_PCC	predicted_positive_voxels	40	4.836e+04	3.024e+04	4.118e+04	2.445e+04
PARTIAL_50	NO_SMOOTHING_PCC	dice_topk	40	0.3886	0.1595	0.3741	0.2825
PARTIAL_50	NO_SMOOTHING_PCC	iou_topk	40	0.2541	0.1352	0.2301	0.1645
PARTIAL_50	NO_SMOOTHING_PCC	dice_fixed05	40	0.2758	0.1662	0.2561	0.1312
PARTIAL_50	NO_SMOOTHING_PCC	iou_fixed05	40	0.172	0.1289	0.1469	0.07022
PARTIAL_50	NO_SMOOTHING_PCC	target_mass	40	1.224e+04	1.333e+04	7200	2489
PARTIAL_50	NO_SMOOTHING_PCC	false_guidance_mass	40	0	0	0	0
PARTIAL_50	NO_SMOOTHING_PCC	outside_clean_target_mas s	40	3.481e+04	2.202e+04	3.027e+04	1.974e+04
PARTIAL_50	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.306e+04	2.792e+04	3.636e+04	2.17e+04
SHIFT_3	FULL_PCC	dice_topk	40	0.3684	0.1713	0.363	0.241
SHIFT_3	FULL_PCC	iou_topk	40	0.2405	0.1442	0.2218	0.1374
SHIFT_3	FULL_PCC	dice_fixed05	40	0.2699	0.1623	0.2635	0.1312
SHIFT_3	FULL_PCC	iou_fixed05	40	0.1672	0.1228	0.1518	0.0702
SHIFT_3	FULL_PCC	target_mass	40	1.334e+04	1.442e+04	7956	2842
SHIFT_3	FULL_PCC	false_guidance_mass	40	1693	1761	962.3	483.2
SHIFT_3	FULL_PCC	outside_clean_target_mas s	40	3.917e+04	2.39e+04	3.363e+04	2.23e+04
SHIFT_3	FULL_PCC	predicted_positive_voxels	40	4.962e+04	3.057e+04	4.254e+04	2.498e+04
SHIFT_3	NO_SMOOTHING_PCC	dice_topk	40	0.38	0.1725	0.3639	0.2525
SHIFT_3	NO_SMOOTHING_PCC	iou_topk	40	0.2498	0.145	0.2225	0.1448

condition	method	metric	N	mean	SD	median	Q1
SHIFT_3	NO_SMOOTHING_PCC	dice_fixed05	40	0.2854	0.1691	0.2809	0.1388
SHIFT_3	NO_SMOOTHING_PCC	iou_fixed05	40	0.1789	0.1303	0.1634	0.07462
SHIFT_3	NO_SMOOTHING_PCC	target_mass	40	1.311e+04	1.418e+04	7843	2784
SHIFT_3	NO_SMOOTHING_PCC	false_guidance_mass	40	2123	2087	1258	625.7
SHIFT_3	NO_SMOOTHING_PCC	outside_clean_target_mas s	40	3.575e+04	2.215e+04	3.062e+04	2.036e+04
SHIFT_3	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.479e+04	2.848e+04	3.759e+04	2.221e+04

Part 2 of 2

Q3	bootstrap 95ci low	bootstrap 95ci high
0.4937	0.3366	0.4419
0.3278	0.2133	0.3047
0.352	0.2279	0.3278
0.2136	0.1357	0.211
1.732e+04	9554	1.835e+04
0	0	0
5.311e+04	3.197e+04	4.659e+04
7.384e+04	4.064e+04	5.935e+04
0.5945	0.4463	0.5555
0.4231	0.3018	0.4077
0.4008	0.2731	0.3829
0.2506	0.1683	0.2559
1.884e+04	1.058e+04	1.968e+04
0	0	0
4.901e+04	2.809e+04	4.18e+04
6.888e+04	3.699e+04	5.415e+04
0.4913	0.3323	0.4386
0.3257	0.2106	0.3011
0.3493	0.2251	0.3241
0.2116	0.134	0.208
1.732e+04	9487	1.842e+04
365.8	178.3	305.7
5.368e+04	3.265e+04	4.724e+04
7.612e+04	4.167e+04	6.023e+04
0.5882	0.4452	0.5527
0.4167	0.3008	0.406
0.3995	0.272	0.3818
0.2496	0.1678	0.2541
1.884e+04	1.052e+04	1.972e+04
771.3	375.7	612.2
4.924e+04	2.851e+04	4.2e+04
6.929e+04	3.718e+04	5.465e+04

Q3	bootstrap 95ci low	bootstrap 95ci high
0.4546	0.2829	0.3849
0.2942	0.1744	0.2562
0.3356	0.2074	0.3053
0.2016	0.1217	0.1965
1.5e+04	8570	1.673e+04
1247	601.9	1058
5.322e+04	3.2e+04	4.695e+04
7.182e+04	4.032e+04	5.884e+04
0.4606	0.2846	0.3882
0.2994	0.1747	0.2583
0.3409	0.2133	0.3152
0.2055	0.1256	0.2029
1.416e+04	8095	1.604e+04
1948	957.7	1615
4.965e+04	2.897e+04	4.233e+04
6.363e+04	3.487e+04	5.218e+04
0.4399	0.2765	0.376
0.282	0.1674	0.2491
0.3253	0.2051	0.3008
0.1943	0.1195	0.1931
1.432e+04	8091	1.603e+04
0	0	0
5.219e+04	3.115e+04	4.571e+04
6.907e+04	3.867e+04	5.717e+04
0.4348	0.2851	0.3822
0.2778	0.1734	0.2526
0.3287	0.2036	0.3032
0.1968	0.1188	0.1965
1.346e+04	7522	1.508e+04
0	0	0
4.901e+04	2.834e+04	4.175e+04
6.065e+04	3.375e+04	5.079e+04
0.4664	0.2976	0.402

Q3	bootstrap 95ci low	bootstrap 95ci high
0.3041	0.1838	0.271
0.3425	0.2128	0.3112
0.2066	0.1258	0.2015
1.523e+04	8604	1.704e+04
0	0	0
5.247e+04	3.142e+04	4.594e+04
7.065e+04	3.944e+04	5.761e+04
0.4878	0.3425	0.4395
0.3226	0.2152	0.2982
0.3508	0.2279	0.3298
0.2128	0.135	0.2142
1.479e+04	8536	1.668e+04
0	0	0
4.901e+04	2.836e+04	4.17e+04
6.371e+04	3.479e+04	5.185e+04
0.4825	0.3167	0.4213
0.318	0.1989	0.2879
0.3477	0.2227	0.3203
0.2105	0.1321	0.2067
1.681e+04	9265	1.795e+04
2498	1197	2268
5.326e+04	3.208e+04	4.676e+04
7.328e+04	4.039e+04	5.907e+04
0.5012	0.3283	0.4346
0.3346	0.207	0.2954
0.3717	0.2353	0.3387
0.2283	0.1411	0.2201
1.665e+04	9081	1.776e+04
3050	1525	2808
5.003e+04	2.926e+04	4.279e+04
6.873e+04	3.631e+04	5.399e+04

Table S8. Internal oracle-assisted controls

Part 1 of 2

method	metric	N	mean	SD	median	min	max
Fixed	topk_Dice	113	0.2683	0.1577	0.2641	4.753e-10	0.6624
Fixed	topk_IoU	113	0.1648	0.1103	0.1522	4.753e-10	0.4952
Naive	topk_Dice	113	0.2686	0.1574	0.2641	4.753e-10	0.6624
Naive	topk_IoU	113	0.165	0.1101	0.1522	4.753e-10	0.4952
EIA-linear	topk_Dice	113	0.3701	0.2284	0.3387	2.193e-09	0.8868
EIA-linear	topk_IoU	113	0.2536	0.1929	0.2039	2.193e-09	0.7966
EIA-blend-0.90	topk_Dice	113	0.3388	0.1771	0.3198	4.753e-10	0.7336
EIA-blend-0.90	topk_IoU	113	0.2181	0.1344	0.1903	4.753e-10	0.5793
EIA-blend-0.75	topk_Dice	113	0.4438	0.2051	0.4128	4.753e-10	0.8854
EIA-blend-0.75	topk_IoU	113	0.3084	0.1797	0.2601	4.753e-10	0.7944
EIA-morph	topk_Dice	113	0.1907	0.1233	0.1777	8.772e-09	0.5808
EIA-morph	topk_IoU	113	0.1109	0.08102	0.09752	8.772e-09	0.4093
Full PCC	topk_Dice	113	0.4398	0.1937	0.4573	5.556e-08	0.8281
Full PCC	topk_IoU	113	0.3013	0.1591	0.2965	5.556e-08	0.7066
No-smoothing PCC	topk_Dice	113	0.6308	0.1958	0.6539	0.05983	1
No-smoothing PCC	topk_IoU	113	0.4893	0.2046	0.4858	0.03084	1

Part 2 of 2

[illegible]

Table S9. RHUH oracle-assisted controls

Part 1 of 2

method	metric	label	n	mean	SD	median	Q1
Fixed	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2066	0.1351	0.2084	0.09385
Fixed	topk IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.1215	0.08608	0.1163	0.04925
Naive	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2066	0.1351	0.2084	0.09385
Naive	topk IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.1215	0.08608	0.1163	0.04925
EIA-linear	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.3435	0.2245	0.2918	0.1657
EIA-linear	topk IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2306	0.1778	0.1708	0.09041
EIA-blend-0.90	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2408	0.1446	0.2619	0.1093
EIA-blend-0.90	topk IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.1444	0.09449	0.1507	0.05783
EIA-blend-0.75	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.346	0.213	0.353	0.1811
EIA-blend-0.75	topk IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2294	0.1627	0.2143	0.09956
Full PCC	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.3612	0.1787	0.3843	0.2382
Full PCC	topk IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2342	0.1309	0.2379	0.1352
No-smoothing PCC	topk_Dice	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.4263	0.1967	0.4292	0.3269

method	metric	label	n	mean	SD	median	Q1
No-smoothing PCC	topk_IoU	ORACLE_ASSISTED_RETROSPECTIVE_LOCALIZATION	39	0.2896	0.1553	0.2733	0.1956

Part 2 of 2

Q3	IQR	bootstrap 95ci low	bootstrap 95ci high	bootstrap replicates	bootstrap seed
0.3039	0.21	0.1651	0.2484	10000	20260810
0.1792	0.1299	0.09516	0.1484	10000	20260810
0.3039	0.21	0.1651	0.2484	10000	20260810
0.1792	0.1299	0.09516	0.1484	10000	20260810
0.527	0.3613	0.2741	0.4134	10000	20260810
0.358	0.2676	0.1764	0.2873	10000	20260810
0.3454	0.2361	0.1963	0.2851	10000	20260810
0.209	0.1511	0.1154	0.1737	10000	20260810
0.5318	0.3507	0.2799	0.4121	10000	20260810
0.3622	0.2626	0.1794	0.2806	10000	20260810
0.4929	0.2547	0.3056	0.414	10000	20260810
0.3271	0.1919	0.194	0.2734	10000	20260810
0.5876	0.2607	0.3651	0.4844	10000	20260810
0.416	0.2204	0.2421	0.3364	10000	20260810

Table S10. Failure accounting

Part 1 of 1

cohort	denominator	failed	status
Development	40	0	complete
Independent internal	113	0	complete
RHUU external	39	0	complete